Anexa 3.1 - CV and track record template (max. 4 pages)

CURRICULUM VITAE

Personal information

Name, Surname:	Fran Nekvapil		
Date of birth:	16. August 1993	Gender:	M
Nationality:	Croatian		
Researcher unique identifier(s) (ORCID, Researcher ID etc.):	UEF-ID (BrainMap): U-1900-061G-9020 ORCID: https://orcid.org/0000-0001-8652-8292		
Personal link from the platform at https://www.brainmap.ro/	https://www.brainmap.ro/fran-nekvapil		
URL for personal website (if case):	https://en.itim-cj.ro/portfolio/nekvapil-fran/		

Education POCU?

Year	Faculty/department - University/institution - Country
2020 (dissertation defended)	Ph.D. in Physics; Babeș-Bolyai University, Faculty of Physics; Romana
2017	Master degree “Mariculture”; University of Dubrovnik, Department of Aquaculture; Croatia

Positions - current and previous
(*Academic sector/research institutes/industrial sector/public sector/other)*

Year	Job title – Employer - Country
2018 - present	R1 First stage researcher (Dec. 2025) – full-time, permanent employment National Institute for Research and Development of Isotopic and Molecular Technologies; research institute; Romania
2020 – present	R2 Recognized researcher (Jul. 2025) – part-time, employment only on projects Babeș-Bolyai University, Ioan Ursu Institute; higher education institution; Romania

Career breaks (if case)

Year	Reason
YYYY-YYYY	

Project management experience
(*Academic sector/research institutes/industrial sector/public sector/other. Please list the most relevant.)*

Year	Project title - Role – Funder – Budget – link to project webpage
2022 - 2023	New composites based on carotenoid-depleted seafood shell waste with efficient environmental pollutants adsorption; Project Leader (PD-type project); funder: UEFISCDI; budget 218 000 RON (cca. 43 500 EUR), 18 months; https://shellpolads.granturi.ubbcluj.ro/
2020 - 2021	Blue bioeconomy approach to wasted marine biomass - an innovative fertilizer for green economy and agriculture (acronym; Project Leader (internal BBU competition); funder Babeș-Bolyai University; budget 50 000 RON (cca. 10 000 EUR), 15 months

Other relevant professional experiences
(e.g. institutional responsibilities, organisation of scientific meetings, membership in academic societies, review boards, advisory boards, committees and major research or innovation collaborations, other commissions of trust in public or private sector)

2021	Main organizer First workshop on Blue Bioeconomy approach to Marine Biomass Waste: Innovative Ideas, Techniques and Translational Sciences *With participation of the company Metrohm Romania, branch B&W TEK* at Babeș-Bolyai University, Cluj-Napoca (Romania) https://bluebiosustain.granturi.ubbcluj.ro/dissem.html#Workshops
2021	Main organizer First workshop on Blue Bioeconomy approach to Marine Biomass Waste: Innovative Ideas, Techniques and Translational Sciences *Reinforcing the collaboration between the Babeș-Bolyai University and the University of Dubrovnik* At University of Dubrovnik, Dubrovnik (Croatia) https://bluebiosustain.granturi.ubbcluj.ro/dissem.html#Workshops

Track record

Original research articles

1. Nekvapil, F., Cardan, I., Cuibus, D., Farcau, C. Nanomolar concentration tracking of the carotenoid astaxanthin by surface-enhanced Raman spectroscopy. *Microchemical Journal*, 2026, 226: 118424; DOI: : 10.1016/j.microc.2026.118424
2. Nekvapil, F., Farcau, C. The role of focused laser plasmonics in shaping SERS spectra of molecules on nanostructured surface. *Nanoscale advances*, 2025, 7: 3008-3017; DOI: 10.1039/d4na00982g.
3. Nekvapil, F., Stefan, M., Marica, I., Falamas, A. A novel observation upon the near-resonant Raman excitation of zinc oxide nanoparticles: Widening of the E2(high) band. *Journal of Raman Spectroscopy*, 2022, 51: 959-968; DOI: 10.1002/jrs.6323.
4. Nekvapil, F., Stegarescu, A., Lung, I., Hirian, R., Cosma, D., Levei, E., Soran, M.-L. A Novel Nanoporous Adsorbent for Pesticides Obtained from Biogenic Calcium Carbonate Derived from Waste Crab Shells. *Nanomaterials*, 2023, 13: 3042; DOI: 10.3390/nano13233042.
5. Nekvapil, F., Bortnic, R-A., Leoștean, C., Barbu-Tudoran, L., Bunge, A. Characterization of the lattice transitions and impurities in manganese and zinc doped ferrite nanoparticles by Raman spectroscopy and x-ray diffraction (XRD). *Analytical Letters*, 2022. 56: 42-52; DOI: 10.1080/00032719.2022.2083145; free
6. Nekvapil, F., Ganea, I-V., Ciorîță, A., Hirian, R., Tomšić, S., Martonos, I.M., Cintă Pinzaru, S. A New Biofertilizer Formulation with Enriched Nutrients Content from Wasted Algal Biomass Extracts Incorporated in Biogenic Powdersl *Sustainability*, 2021, 13: 8777; DOI: 10.3390/su13168777
7. Nekvapil, F., Cintă Pinzaru, S., Barbu-Tudoran, L., Suciu, M., Tamas, T., Chis, V. Color-specific porosity in double pigmented natural 3d-nanoarchitectures of blue crab shell. *Scientific Reports*, 2020, 10: 3019; DOI: 10.1038/s41598-020-60031-4.
8. Nekvapil, F., Aluas, M., Barbu-Tudoran, L., Suciu, M., Bortnic, R-A, Glamuzina, B., Cintă Pinzaru, S. From Blue Bioeconomy toward Circular Economy through High-Sensitivity Analytical Research on Waste Blue Crab Shells. *ACS Sustainable Chemistry & Engineering*, 2019, 7: 16820-16827; DOI: 10.1021/acssuschemeng.9b04362

Patent

- Nekvapil, F., Soran, M-L., Stegarescu, A. Material adsrobant pentru poluanți obținut din deșeuri de cochilii de crab – metoda de obținere; Publicat in baza de date *Espacenet*, code RO138758A2 (URL: https://worldwide.espacenet.com/patent/search/family/099821365/publication/RO138758A2?q=Fran%20Nekvapil)

Internal INCDTIM Method file

- Determinarea distribuției ocupării siturilor în nanoparticulele de oxid de fier utilizând spectroscopia Raman; INCDTIM registration nr. 2396/5.05.2025

Narrative CV

A narrative summarizing which work has had the greatest importance and impact.

The interdisciplinary involvement of dr. Nekvapil began during his Master study cycle at the University of Dubrovnik. He studied “mariculture”, a science of cultivating aquatic organisms for human consumption and pharma industry, however, in the year 2015 he came into contact with his future Ph.D. mentor, Prof. Cîntă Pinzaru from Babeș-Bolyai University (BBU), and became interested in high-precision spectroscopy analytical methods.

This occasional contact with world of physics developed into a collaboration where his Master thesis presented the applications of Raman spectroscopy in studies of marine organisms and chemical composition of their tissues. Further, he moved to the Faculty of Physics at the BBU, where he defended his doctoral thesis titled “Raman Spectroscopy Techniques and Applications to Carotenoids Research in Marine Living Resources and to the Blue Bioeconomy” in 2020.

Dr. Nekvapil has continued with the research of aquatic resources using high-precision analytical methods (Raman spectroscopy techniques, XRD, electron microscopy, etc.) and waste management. In 2018 he got employed at the INCDTIM Cluj, where he embarked onto another research direction concerning nanomaterials engineering, relying on his profound knowledge of spectroscopy analytical techniques. In 2021, he joined the Nanophotonix team at INCDTIM, whereafter he reduced the intensity of waste management research and instead focused more on investigations of fundamental spectroscopy processes and plasmonic mechanisms.

At INCDTIM, he worked as a team member at the project SANOMAT (https://infim.ro/en/project/sanomat/) where he used Raman spectroscopy in concert with XRD, XPS magnetization to establish novel structure-signal correlation, namely the correlation of the Raman signal of cation vacancies and saturation magnetization of respective Mn-Zn ferrite nanoparticles. Dr. Nekvapil immediately continued working as a team member of the project ZnOFluoSERS (https://sites.google.com/view/znofluosers/home) which aimed to prospect the plasmonic activity of zinc oxide nanoparticles and their suitability for Surface-enhanced Raman spectroscopy and Surface-enhanced fluorescence. Here he reported on multi-laser Raman excitation and signal configuration of ZnO nanoparticles of different sizes and synthesized by different methods.

- Outside of projects, dr. Nekvapil continued to apply spectroscopy techniques for structural characterization of various ferrites, including magnetite, Co-ferrites, Zn-ferrites, Mn-ferrites and others. This resulted in accumulation of significant spectral database and unique insights into structure-signal relationships. In an attempt to formalize these signal interpretation insights in a written form and facilitate the learning curve for newcomers to the field, he created a reference framework which is currently reported as an internal method file titled “Determining the cation site occupancy rates in iron oxide nanoparticles using Raman spectroscopy”. This framework is a work still in progress and will be updated periodically as new data are obtained.

In parallel, dr. Nekvapil conducted a Post-doc type project Shell-Pol-Ads (https://shellpolads.granturi.ubbcluj.ro/), where he sought to exploit the nanoporous character of wasted crab shells and patented a method for obtaining a nature-derived adsorbent material which was tested against pesticides and antibiotics. Currently, dr. Nekvapil is looking to connect more tightly his research on plasmonics and that of magnetic ferrite nanomaterials, in the context of environment decontamination.

Admittedly, dr. Nekvapil has achieved a lot and has a quite active research career despite his handicap of severely impaired vision. He worked with impermanent contract arrangements for several years, until finally obtaining a permanent position in the Nanophotonix team at INCDTIM Cluj.

The first prooject dr. Nekvapil led (MARIFERT, ref. X) was an internal grant of the BBU. Nevertheless, this grant presumed an evaluated and approved research plan, scientific reporting and financial management. Scientific coordination of several colleagues was also necessary. Subsequently, he managed a post-doc project at BBU (POCU) introduced in the previous paragraph, which was a solo-stand. In the same time, the PL led a Post doc project awarded through the national competition (PD 46/2022; Shell-Pol-Ads). This type of project assumes a project leader who works under the supervision of a scientific mentor, hence also a project without a formal team. Hence, dr. Nekvapil has certain experience in managing projects and managing an informal team.